Description

Illustration





Figure caption

1. Close up of glulam showing layers of wood. 2. Glulam support beams within the Art Gallery of Ontario. Images used under license from Shutterstock.com

The material

Glulam (glued-laminated timber) is a structural timber product manufactured by gluing together individual pieces of dimension lumber under controlled conditions. It is widely used as an attractive architectural and structural building material. The grade-specification depends on whether it is for a beam, a column or a tension member. For the bending grades of 20f-E, 20f-EX, 24f-E and 24f-EX, the numbers 20 and 24 are the allowable bending stress in Imperial units (2000 and 2400 pounds per square inch). The f refers to flexure and E indicates that most laminations must be tested for stiffness. Grades with EX designation (20f-EX and 24f-EX) are used when bending members are subjected to stress reversals. In these members the lamination requirements in the tension side are the mirror image of those in the compression side. Similarly the descriptions for compression grades,16c-E and 12c-E, and tension grades,18t-E and 14t-E indicate the allowable compression and tension stresses.
Glulam is manufactured in three grades: Industrial, Commercial and Quality. Industrial Grade is used when appearance is not a primary concern. It may have broken knots, knot holes, torn grain, checks, and other irregularities. Commercial Grade is intended for painted or flat-gloss varnished surfaces. It may have knot holes, loose knots and voids, not replaced by wood inserts or filler on exposed surfaces.
Quality Grade is for high-gloss transparent or polished surfaces, displaying the natural beauty of the wood. Split knots, slivers, torn grain or checks on the surface are filled; loose knots, knot holes, wane and pitch pockets removed and replaced with wood inserts of matching grain.

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O

General properties

Density		500	-	650	kg/m^3
Price	*	1.33	-	1.99	GBP/kg

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus		12	-	14	GPa
Shear modulus		1	-	2	GPa
Bulk modulus		8	-	10	GPa
Bending modulus	*	12	-	14	GPa
Poisson's ratio		0.25	-	0.3	
Yield strength (elastic limit)		9	-	12	MPa
Tensile strength		9.65	-	12.4	MPa
Compressive strength	*	8.3	-	11	MPa
Bending strength		13.8	-	16.5	MPa
Elongation		1.5	-	2.5	% strain
Hardness - Vickers		3	-	4	HV
Fatigue strength at 10^7 cycles	*	7.3	-	9.8	MPa
Fracture toughness		4	-	6	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.008	-	0.01	

Thermal and combustion properties

Thermal conductor or insulator?		Good insulator			
Thermal resistivity		2.63	-	3.33	m.°C/W
Thermal expansion coefficient		3	-	10	µstrain/°C
Specific heat capacity		1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature		120	-	140	°C
Minimum service temperature		-73	-	-23	°C
Flammability		Highly flammable			
Emissivity	*	0.81	-	0.9	

Hygro-thermal properties

Water absorption		12	-	25	%
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	1e13	-	1e14	µohm.cm
Dielectric constant (relative permittivity)	5.5	-	6.5	
Dissipation factor (dielectric loss tangent)	0.06	-	0.75	
Dielectric strength (dielectric breakdown)	0.4	-	0.6	MV/m

Optical properties

Natural color

Golden brown

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4
Machinability	4	-	5

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Limited use
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Acceptable
Rural atmosphere	Acceptable
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.435	-	0.48	kg/kg
Embodied energy (virgin grade production)	* 41.9	-	46.1	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 15.6	-	17.2	MJ/kg
Machining, fine, embodied energy	* 17.5	-	19.4	MJ/kg
Machining, grinding, embodied energy	* 19.4	-	21.5	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 1.02	-	1.13	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.15	-	1.27	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.27	-	1.41	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.226	-	0.25	kg/kg
Embodied energy, recycling	* 34.8	-	38.3	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.5	-	20.5	MJ/kg
Combustion CO2	* 1.73	-	1.82	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Design guidelines

Laminating is an effective way of using high strength lumber of limited dimension to make large structural members in many shapes and sizes. Glulam is used for columns and for beams and for curved members loaded in combined bending and compression. It costs about twice as much as the wood from which it is made - the extra cost is that of the glue, the labor and overheads.

Technical notes

In the manufacture of Glulam, the wood pieces are end jointed and arranged in horizontal layers or laminations. The lumber used for its manufacture is a special grade (lamstock) which is purchased directly from lumber mills. It is dried to a maximum moisture content of 15 percent and it is planed to a closer tolerance than that required for dimension lumber. Canadian Glulam is manufactured in three species combinations: Douglas Fir-Larch, Hem-Fir and Spruce-Pine.

Values marked * are estimates.
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Typical uses

Glulam is used for large-span load-bearing structures, both concealed and visible: roof beams, both straight and curved, floor beams, columns and trusses.

Related standards

CSI/CSC Masterformat Section Numbers
06 18 00 Glued-Laminated Construction

Further reading

Fridley, K.J. (2002) Wood and Wood-Based Materials: Current Status and Future of a Structural Material. American Society of Civil Engineers, Journal of Materials in Civil Engineering, Vol.14, No.2: pp. 91-96.

Links

Reference

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